

WHAT DO INTERACTION NETWORK METRICS TELL US ABOUT SPECIALIZATION AND BIOLOGICAL TRAITS?

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Abstract. The structure of ecological interaction networks is often interpreted as a product of meaningful ecological and evolutionary mechanisms that shape the degree of specialization in community associations. However, here we show that both unweighted network metrics (connectance, nestedness, and degree distribution) and weighted network metrics (interaction evenness, interaction strength asymmetry) are strongly constrained and biased by the number of observations. Rarely observed species are inevitably regarded as “specialists,” irrespective of their actual associations, leading to biased estimates of specialization. Consequently, a skewed distribution of species observation records (such as the lognormal), combined with a relatively low sampling density typical for ecological data, already generates a “nested” and poorly “connected” network with “asymmetric interaction strengths” when interactions are neutral. This is confirmed by null model simulations of bipartite networks, assuming that partners associate randomly in the absence of any specialization and any variation in the correspondence of biological traits between associated species (trait matching). Variation in the skewness of the frequency distribution fundamentally changes the outcome of network metrics. Therefore, interpretation of network metrics in terms of fundamental specialization and trait matching requires an appropriate control for such severe constraints imposed by information deficits. When using an alternative approach that controls for these effects, most natural networks of mutualistic or antagonistic systems show a significantly higher degree of reciprocal specialization (exclusiveness) than expected under neutral conditions. A higher exclusiveness is coherent with a tighter coevolution and suggests a lower ecological redundancy than implied by nested networks.

Key words: abundance distribution; biological traits; connectance; degree distribution; ecological networks; interaction diversity; interaction strength; nestedness; null models; specialization.

INTRODUCTION

The analysis of ecological interaction networks is an important tool in community ecology and is often used to draw conclusions about such diverse topics as community robustness, biodiversity maintenance, resource partitioning, and natural selection (May 1972, Pimm 1982, Pimm et al. 1991, McCann et al. 1998, Proulx et al. 2005, Bascompte et al. 2006, Beckerman et al. 2006, Montoya et al. 2006, Santamaría and Rodríguez-Gironés 2007). Ecological interaction networks are representations of associations (links) between species (nodes). Network studies like those just cited commonly select one or a few network metrics (Table 1) that are based either on unweighted links (only representing the presence or absence of a link, also known as “qualitative” or binary metrics) or on weighted links (“quantitative” metrics). Important unweighted metrics include connectance (May 1972,

Jordano 1987, Beckerman et al. 2006, Santamaría and Rodríguez-Gironés 2007), nestedness (Bascompte et al. 2003, Santamaría and Rodríguez-Gironés 2007), and degree distribution (Jordano et al. 2003, Vázquez 2005). Among weighted metrics, dependence or interaction strength (Bascompte et al. 2006, Vázquez et al. 2007) or different measures of interaction diversity or evenness (Bersier et al. 2002, Tylianakis et al. 2007) are commonly used.

Describing network patterns based on these metrics may help to unravel important patterns of community organization and heterogeneity in trophic associations, particularly if the underlying data represent an unbiased estimate of realized interactions. However, network patterns are often interpreted as a result of biological trait matching and fundamental specialization or generalization, which reflects the dependency between species or redundancy of associations, respectively, and thus contributes to the stable coexistence of species in communities (Pimm et al. 1991, Montoya et al. 2006). For instance, secondary extinctions of consumers due to resource losses have been suggested to decrease with increasing connectance (Dunne et al. 2002b, Estrada 2007), whereas nestedness and asymmetries in interaction strength may promote community stability (Bas-

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TABLE 1. Definitions of common bipartite network metrics.

Metric	Definition
A) Metrics based on unweighted links	
Connectance	The proportion of possible links actually observed in a web. The connectance (or connectivity) is: $C = L/(IJ)$ for bipartite networks, or $C = L/S^2$ in unipartite networks (e.g., food webs), respectively. L describes the number of realized links; I and J are the number of species of each party in bipartite networks, e.g., hosts vs. parasites; S is the total number of species. The related “linkage density” is $D = L/(I + J)$.
Generality and vulnerability	In a network of I consumers and J prey species, the mean number of prey species (links) per consumer is termed “generality” ($G = L/I$) and the mean links per prey “vulnerability”: ($V = L/J$) (Schoener 1989).
Nestedness	The nestedness “temperature” T (0° – 100°) (Atmar and Patterson 1993) measures the departure from a perfectly nested interaction matrix (degree of disorder). $T = 0^\circ$ is defined for maximum nestedness: when rows and columns are ordered by decreasing number of links, links of each row and column exactly represent a subset of the previous ones. For each species, the deviation from the expected nested order accounts for a higher “idiosyncratic temperature” (Atmar and Patterson 1993). Nestedness can be defined as $N = (100^\circ - T)/100^\circ$ (Bascompte et al. 2003).
Degree distribution	For each species, the number of links describes its “degree” k . A power law or “scale-free” distribution implies that the cumulative degree distribution (i.e., the proportion of species with k or more links) is described by $P(k) \sim k^{-\gamma}$, where γ is a constant. Many ecological networks are better fitted by a truncated power law function $P(k) \sim k^{-\gamma} \exp(-k/k_x)$, assuming an exponential decay of the power law distribution, where k_x represents an additional constant (Jordano et al. 2003, Vázquez 2005).
B) Metrics based on weighted links†	
Interaction strength	Interaction strength of species j on species i (b_{ij}) can be defined by the proportion of interactions between i and j (a_{ij}) of the total interactions recorded for i ; thus $b_{ij} = a_{ij}/\sum_{j=1}^J a_{ij}$. For mutualistic networks, Jordano (1987) and Bascompte et al. (2006) used b_{ij} as a measure of dependence of species i on its partner j . Asymmetries of interaction strength can be defined as $AS_{ij} = (b_{ij} - b_{ji})/(b_{ij} + b_{ji})$, where b_{ji} is the reciprocal dependence of species j on species i (see Bascompte et al. 2006, Blüthgen et al. 2007, Vázquez et al. 2007).
Interaction diversity	The Shannon diversity of links is $H_i = -\sum_{j=1}^J [(a_{ij}/A_i) \ln(a_{ij}/A_i)]$ for species i , or for the whole web, $H_2 = -\sum_{i=1}^I \sum_{j=1}^J [(a_{ij}/m) \ln(a_{ij}/m)]$, with $A_i = \sum_{j=1}^J a_{ij}$ and $m = \sum_{i=1}^I \sum_{j=1}^J a_{ij}$. Their reciprocals e^{H_i} and e^{H_2} express the equivalent “effective” number of links (see Bersier et al. 2002).
Interaction evenness	Based on Shannon diversity, the interaction evenness is $E_i = H_i/\ln L_i$ for each species, or for the whole web, $E_2 = H_2/\ln L$, where L_i is the number of links of species i , and L is the number of all links. First suggested by Bersier et al. (2002), these measures or other standard diversity metrics have been applied to different interaction networks (e.g., Sahli and Conner 2006, Albrecht et al. 2007, Tylianakis et al. 2007).
Weighted generality and vulnerability	The weighted analog of generality can be derived from Shannon diversity of links (H_i), representing the mean “effective” links per consumer $G_i = (1/I) \sum_{j=1}^J e^{H_{ij}}$, or as the weighted mean $G_{qw} = \sum_{j=1}^J (A_i/m) e^{H_{ij}}$ (Bersier et al. 2002, with equations based on $\log_2$ instead of $\ln$). For weighted vulnerability, replace i by j and I by J .
Standardized interaction diversity	Derived from Shannon diversity of links in the network (H_2), the following specialization index (H'_2 ; Blüthgen et al. 2006) increases with the deviation of realized interaction frequencies from expected values of a null distribution of interactions (H_{2min} , corresponding to a perfectly quantitatively “nested” matrix) based on the given species frequency distribution (fixed interaction totals): $H'_2 = (H_{2max} - H_2)/(H_{2max} - H_{2min})$, where H_2 is defined as above. Note that a χ^2 analysis of homogeneity is conceptually similar.
Standardized distance	In a similar way, the nonconformity of a focal species i can be described in relation to total interaction frequencies (standardized Kullback-Leibler distance d'_i , Blüthgen et al. 2006). In analogy to “idiosyncratic temperatures” defined for nestedness above, d'_i indicates the exclusiveness of the interactions of a species, i.e., its deviation from a null distribution assuming that interactions of i represent a subset of the overall sample. Consequently, d'_i increases with reciprocal specialization between i and its partner.

† Link weight is often defined as interaction frequency (a_{ij}) between species i and species j from a total set of I vs. J species. The sum of all interaction for species i is $A_i = \sum_{j=1}^J a_{ij}$, for j $A_j = \sum_{i=1}^I a_{ij}$, and for the whole network $m = \sum_{i=1}^I \sum_{j=1}^J a_{ij}$. Most equations in the section “Metrics based on weighted links” are only valid if species lacking any interaction ($A_i = 0$ or $A_j = 0$) are excluded.

compte et al. 2003, 2006). In the present paper, we will argue that these raw metrics as such, uncontrolled for neutrality, may be substantially flawed regarding an interpretation of trait matching or fundamental specialization. In this study we define *trait matching* as the degree of interaction partitioning between species, resulting from the correspondence of phenotypic traits of interacting species (Fig. 1); these traits include body size, morphology, chemical composition, physiological abilities, temporal activity patterns, preferences, and

behavior. For instance, species-specific preferences may strengthen some links, while structural barriers or an avoidance of defenses may inhibit others. A strong structural importance of trait matching leads to a high level of *specialization* in the community (“fundamental specialization” sensu Vázquez and Aizen 2006), while the absence of structuring by trait matching implies maximum *generalization*. The latter scenario is often simulated in null model networks, where partners associate randomly (Vázquez and Aizen 2006).

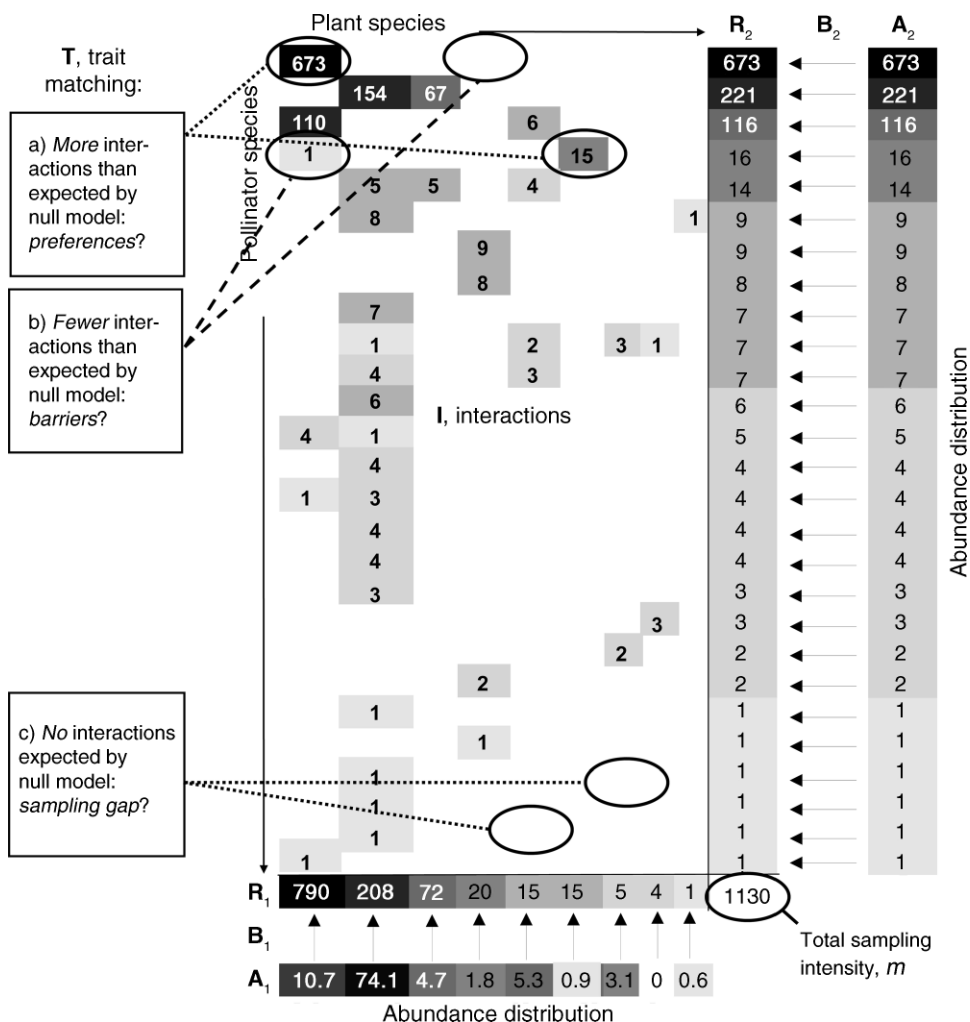


FIG. 1. Conceptual framework of a quantitative interaction network, represented by an empirical pollination network of nine plant species and 27 pollinator species ("Safariland" in Vázquez and Simberloff 2003). The different levels are: abundance distribution (A_1 and A_2 for plants and pollinators, respectively), total observation records per species (R_1 and R_2), and the distribution of interactions (I) in the core of the network. Numbers in the boxes are the number of observed interactions between a particular plant species and a particular pollinator species: $R_1 = 790$ indicates that there were 790 observations of interactions between one plant species and six different pollinator species, and $R_2 = 673$ indicates that there were 673 observations of one pollinator species at a single plant species. The darkness of the shading increases with the number of observations and serves to show the pattern of links in the matrix. General biological traits (B_1 and B_2), overall sampling intensity, and sampling bias may affect the relationship between A and R . The interactions (I) are shaped by R and by trait matching between associated partners (T). The effect caused by T can be studied by comparing the distribution of neutral encounter probabilities based on R (null model) with the actual distribution of interactions (I). This comparison may facilitate the search for biological processes (examples a–c), while the scenarios (b) and (c) are not distinguished in network metrics when predictions based on R are uncontrolled for. Note that A_1 represents an independent estimate of plant densities (mean number of individuals per 20-m² quadrat). Such independent measures are unavailable for pollinators in this network; hence, A_2 represents the number of individual pollinators observed on flowers ($A_2 = R_2$). The prediction based on the null model is displayed in Fig 2b.

In addition to trait matching or neutral interactions, however, metrics depicting the network structure may also be affected by variation in sampling intensity (Paine 1988, Vázquez and Aizen 2006). This impact was studied using rarefaction techniques (Goldwasser and Roughgarden 1997, Banašek-Richter et al. 2004, Herrera 2005, Blüthgen et al. 2006). It is a well-known fact that ecological data are constructed of incomplete samples: only a subset of the interactions naturally present in a

community is recorded. However, the consequences are often ignored when these metrics are currently used for characterizing specialization in ecological networks. Most importantly, for rarely observed species (e.g., see Plate 1) most metrics are necessarily biased, a simple fact that is poorly considered in interpreting networks. For example, species that are represented only by a single observation (singletons) are inevitably assigned only a single link and regarded as "specialists," inflating the

estimate of the prevalence of specialization, nestedness, and strength asymmetry (if singletons are not associated with exclusive partners), and deflating connectance and interaction diversity. Because ecological networks usually include several rarely collected species with limited observation records (including singletons; see Fig. 1), it is crucial to know to what extent network patterns can be attributed to biological processes such as trait matching, in contrast to effects of sampling intensity, sampling bias, or the underlying species abundances.

We therefore examined the sensitivity of commonly used network metrics for variation in the number of observations per species, given a realistic level of overall sampling intensity and thus average observations per species. In most ecological data sets, some species will be much more commonly recorded than others, a consequence of the abundance distribution that is typically log-normal or alike (McGill et al. 2007), different activity patterns, and possibly sampling bias. A typical network thus involves a mixture of species at different information levels, and the relative contribution of frequently vs. poorly observed species may vary across networks. Such variation in observation frequency per species affects the detectability of links in unweighted metrics, but also the relative strength of links in weighted metrics. This combined impact of rarely and frequently recorded species may need to be considered in network comparisons, particularly if networks differ in the shape of the frequency distribution.

The goals of the present study are threefold: (1) to compare the degree of skewness of the frequency distribution of observations per species across a set of empirical networks; (2) to use simulations to evaluate how much this skewness affects the metrics in the absence of trait matching and fundamental specialization; and (3) to compare the biological conclusions drawn from earlier network metrics with those derived from an information-theoretical approach (Blüthgen et al. 2006) that controls for variable number of observations. We focused on bipartite networks, assuming that all species of one party can potentially interact with all partners of the other party and are completely opportunistic in their associations and consequently interact in the most generalized way possible. This generalized scenario was represented by simulating neutral interactions between species in a null model approach. If a link is absent in such a simulated network, we thus assume that this is caused by the information deficit due to limited sampling. Whereas some previous studies also modeled the effect of the specific abundance or interaction frequency on individual network metrics (Vázquez and Aizen 2003, Vázquez 2005, Vázquez et al. 2005, 2007), we explicitly varied only the skewness of the frequency distribution in the null model to examine its structural consequences in the absence of other variables, and compared its effect across all commonly used metrics for the first time.

SIMULATION METHODS

A bipartite ecological network describes the interactions between two communities containing I and J species. Networks can be displayed as $I \times J$ contingency tables, where each cell entry depicts the number of interactions recorded between a specific pair of species (Fig. 1). Our model of neutral encounter probabilities is based on the marginal totals of this contingency table, defining the total number of observation records per species as the vectors $\mathbf{R}_1$ and $\mathbf{R}_2$ (Fig. 1). Henceforth, we refer to $\mathbf{R}_1$ and $\mathbf{R}_2$ together as $\mathbf{R}_k$ (for $k = [1, 2]$).

Null models are useful to evaluate whether structural patterns may be produced by stochastic processes in the absence of particular mechanisms (Gotelli and Graves 1996). In this case, we aimed to investigate how network metrics are determined by the frequency distribution in the total observation records per species, in the absence of any limitation by trait matching. We assume that the observation records per species ($\mathbf{R}_k$) in natural networks are driven mainly by sampling intensity, sampling bias, species abundances, or variation in general species traits that are independent of the specific distribution of the interactions *between* the parties I and J (Fig. 1). The null model scenario thus implies that specialization and abundance vary independently and do not interact, thereby ignoring potential feedbacks from the interaction level that might influence $\mathbf{R}_k$ as emergent network property. For given marginal totals ($\mathbf{R}_k$), the expected value for the interactions between species i and species j can be calculated as $A_i A_j / m$, where A_i and A_j represent the total observation records of species i and j , respectively, and m is the grand total number of observed interactions in the network. This procedure is analogous to calculating expected values in a chi square test of homogeneity. Note that this theoretical expectation does not constrain interactions to integer values and does not provide stochastic variation of interactions. The null model algorithm applied here (Patefield 1981) uses fixed marginal totals to distribute the interactions and produce a set of networks where all species are randomly associated. Patefield's (1981) algorithm is implemented as function "r2dtable(100, $\mathbf{R}_1$, $\mathbf{R}_2$)" in R statistical software version 2.4.1 (R Development Core Team 2006). We generated 100 randomizations for each specific distribution of marginal totals. An example of a null model web generated from $\mathbf{R}_k$ of a plant–pollinator network (Fig. 1) is displayed in Fig. 2b.

In addition to Patefield's (1981) algorithm, two alternative null models were used, in which the marginal totals $\mathbf{R}_k$ defined the probability of an interaction but were not fixed. In the first alternative model, at least one link was assigned to each species in order to maintain the network size (as in Vázquez et al. 2007, but without fixing connectance). This condition to maintain one link per species was not implemented in the second alternative null model, allowing the network to become smaller when some species were assigned no interactions and hence disappeared from the network (Fig. 3c).

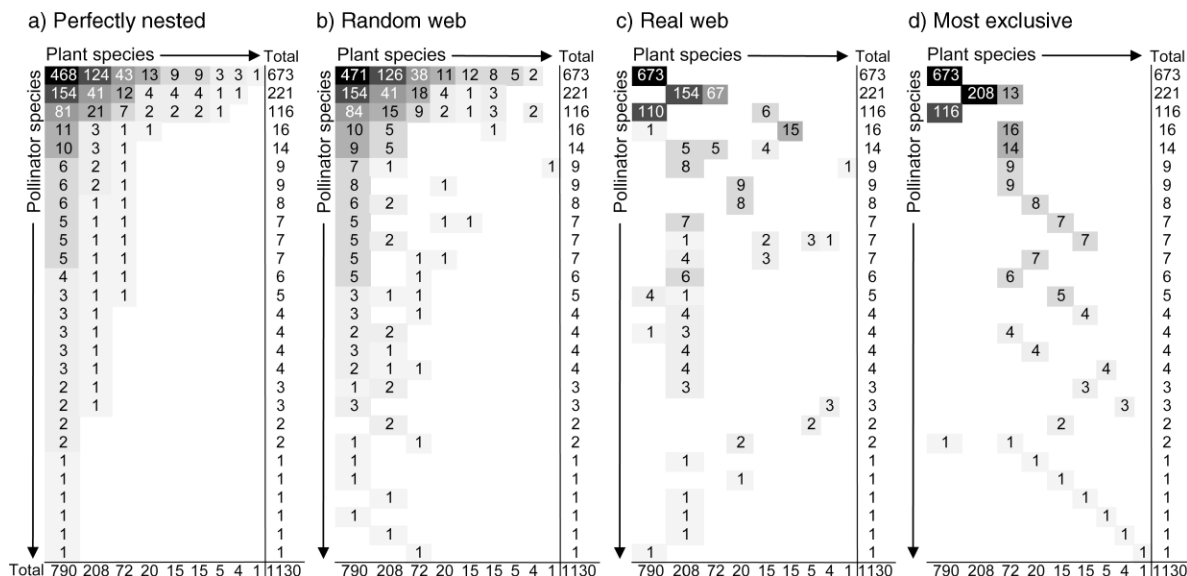


FIG. 2. Four networks with the same distribution of total observation records ($\Rightarrow$ row and column totals) of 27 pollinator and nine plant species, but increasing exclusiveness of interactions. Numbers and shading are as described for Fig. 1. (a) Hypothetical network showing a perfectly nested distribution of interactions ($H'_2 = 0.0$). (b) One network generated from randomized interactions ($H'_2 = 0.03$). (c) An empirical pollination network (as in Fig. 1; $H'_2 = 0.85$). (d) A hypothetical network with the most exclusive distribution of interactions possible for the same frequency distribution ($H'_2 = 1.0$). Randomized interactions as in (b) yield an average $H'_{2ran} = 0.016 \pm 0.007$ (mean $\pm$ SD of 100 simulations).

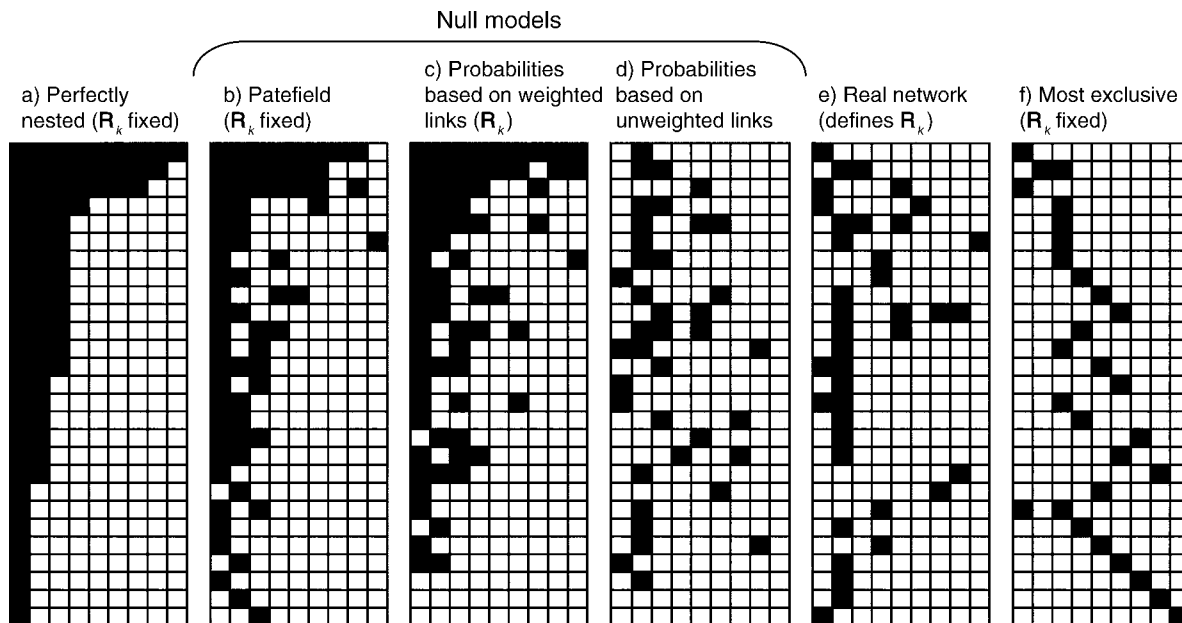


FIG. 3. Networks (27 pollinator $\times$ 9 plant species) displayed in their unweighted form, with links coded as present (black) or absent (white). The empirical network (e), the extreme scenarios (a, f), and a network based on Patefield's (1981) algorithm (b) are equivalent to the four webs shown in Fig. 2 and maintain the species observation records R_k from (e). Networks derived from two additional null models are shown: for (c), R_k defined the probability of an interaction, but was not fixed; network (d) was generated with the null model "CE" previously established for nestedness analysis based on the number of links. Note that (b) and (c) more closely resemble the most nested scenario, whereas (d) strongly deviates from perfect nestedness based on R_k . The original order of rows and columns was maintained as in Fig. 2, except for (c) and (d), where some pollinator species and one plant species did not receive a link and were rearranged.



PLATE 1. Many insect species are rare, rarely observed, or only occasionally forage on flowers such as this scorpion fly. Conventional network metrics regard such flower visitors as “specialized”—they are assigned only one or few links and high values of interaction strength or dependence. They also contribute to low connectance and to a nested pattern of a network, if they interact with commonly visited flowers, e.g., a hogweed shown here. Photo credit: Michael Werner.

Because the three null models revealed very similar results, only those generated by the Patefield (1981) algorithm are presented in detail, and results based on the least conservative third null model are shown in the Appendix.

In order to manipulate the skewness of $\mathbf{R}_k$, we varied the standard deviation (σ) of a log normal distribution, while maintaining the grand total number of observations (m). Both $\mathbf{R}_1$ and $\mathbf{R}_2$ were generated by drawing series of I or J random numbers from a log normal distribution, with $\mu = \ln(M) - 0.5 \sigma^2$, where μ is the mean, σ is the standard deviation, and M describes the sampling density of the network (mean number of observed interactions per cell), obtained from the grand total number of interactions recorded (m) as $M = m/(IJ)$. Each $\mathbf{R}_k$ was the median distribution of 100 sorted random drawings. Variation in σ generated a different heterogeneity of $\mathbf{R}_k$. We chose σ values ranging from 0 to 10, and calculated μ accordingly. In order to quantify the heterogeneity of $\mathbf{R}_1$, E_{R1} , we used the evenness index based on Shannon diversity:

$$E_{R1} = \frac{\sum_{i=1}^I p_i \ln p_i}{\ln I}$$

where p_i is the proportion of the interaction totals for species i (A_i) of all interactions in the community (m); hence, $p_i = A_i/m$. The heterogeneity of $\mathbf{R}_2$ (E_{R2}) was

defined accordingly (replace i by j and I by J). Evenness approaches 0 for the most heterogeneous distribution and 1 for a perfectly homogenous distribution. $\mathbf{R}_1$ and/or $\mathbf{R}_2$ were often slightly readjusted in order to allow the evenness for both parties to become as similar as possible ($E_{R1} \approx E_{R2}$) and to ensure that $\Sigma(\mathbf{R}_1) = \Sigma(\mathbf{R}_2) = m$.

We generated networks of a size and interaction density typical for empirical studies. All networks shown here thus contained $I \times J = 30 \times 10$ species and a total of $m = 600$ interactions, equivalent to an average of two interactions per cell, $m/(IJ) = 2$. These values for network size and interaction density are similar to the median found across 51 empirical mutualistic networks (26×10 and 2.1, respectively; compiled in Blüthgen et al. 2007). Other network sizes and shapes, e.g., squared networks ($I = J$), were examined as well, but general trends remained largely unaffected (see Appendix).

To calculate nestedness temperature (Table 1: Nestedness), we used the program Aninhado 2.0.2 (Guimarães and Guimarães 2006), which complements the original procedure by Atmar and Patterson (1993) with improved null models and allows an efficient calculation of a large number of matrices. We also compared our null model simulations to the results of significant nestedness inferred from a previously used null model based on unweighted networks (null model “CE” in Aninhado 2.0.2, among other studies used in Bascompte et al. [2003], Guimarães et al. [2006, 2007]). This null model

assigns a link between each pair of species with a probability deduced from the product of their total number of links (species degree). This null model has been considered conservative for nestedness analysis compared to alternative unweighted null models (Bascompte et al. 2003, Guimarães et al. 2006). A typical CE null model web is shown in Fig. 3d.

We characterized the degree distribution (Jordano et al. 2003) by fitting the observed distribution to two models, the power law (PL), and the truncated power law (TPL) (Table 1: Degree distribution). Because TPL is an extension of PL and requires an additional parameter, namely a constant defining the exponential decay, the Akaike Information Criterion (AIC) was used to evaluate which model provides the better fit. TPL was considered superior to PL only if the difference in AIC (i.e., ΔAIC) was greater than 2.

RESULTS AND DISCUSSION

Variation in natural networks

In order to maintain a constant sampling density $M = m/(IJ)$ across different networks, the overall number of observations (m) must increase proportionally to the size of each network (IJ). The slope is clearly lower for empirical data sets. We analyzed 51 mutualistic plant–animal networks, including pollinators, seed dispersers, and ants from six continents (Blüthgen et al. 2007): many of them are *available online*.⁵ All networks comprised weighted links, represented by the number of individuals of an animal species recorded on a particular plant species. The networks differed strongly in the number of species and observations (range 3–219 plant species, 4–679 animal species; $39 \leq m \leq 19946$). Whereas m increases consistently with network size (IJ) across the 51 networks (Spearman's $r_s = 0.65$, $P < 0.0001$), the increase is weak and much less than proportional to IJ , since M declines monotonically over IJ ($r_s = -0.57$, $P < 0.0001$). Therefore, larger networks typically contain fewer observations per potential link.

Total interaction frequencies per species in different assemblages vary substantially in their heterogeneity, causing a high variation in the skewness of marginal totals ($\mathbf{R}_k$) in different networks. $\mathbf{R}_k$ vectors of plants and animals in the set of 51 mutualistic networks cover a broad range of evenness values, ranging from 0.36 to 1.00 (mean $\pm$ SD: 0.75 ± 0.13 ; evenness based on Shannon diversity). Particularly homogenous $\mathbf{R}_k$ vectors were recorded for myrmecophytic plants (0.88 ± 0.09 , $n = 14$ networks), while pollinators, on average, had the lowest evenness (0.67 ± 0.14 , $n = 21$ networks), but variation was high within each network type as well. Given this broad variability of the skewness of $\mathbf{R}_k$, network comparisons merit a closer examination of the impact of $\mathbf{R}_k$ alone.

Connectance

Connectance (C) is one of the simplest and most commonly used metrics to describe the density of links in interaction networks or food webs (Table 1), and is usually interpreted as the degree of generalization or redundancy in a system, with consequences for community stability (May 1972, Dunne et al. 2002b, Estrada 2007). However, sampling intensity is crucial. In a scenario where all species are completely generalized and interactions are unlimited by trait matching, additional sampling would increase the number of links between these species and thus connectance, eventually leading to a fully connected network. Therefore, in the absence of sampling limitation, the expected connectance would be $C = 1$. In our null-model analysis of generalized associations based on limited sampling intensity, connectance was much lower than 1 in networks with few frequent and many rare species, and showed a substantial decline with increasing skewness of the frequency distribution ($\mathbf{R}_k$) (Fig. 4a). Therefore, variation in the species sampling frequencies, or in the underlying abundance distribution, additionally shapes C apart from the previously documented effect of total species numbers or total sampling effort (Goldwasser and Roughgarden 1997, Olesen and Jordano 2002, Banašek-Richter et al. 2004, Blüthgen et al. 2006).

In addition, an increased information deficit in larger networks may, at least partly, account for the hyperbolic decline in C over network size. If the same absolute effort is taken to sample larger networks, the number of observations per species will decline with increasing number of possible links (IJ), and so will C (see Kenny and Loehle 1991). Alternative explanations for this decay have been proposed, including an increased specialization in larger networks, “forbidden links,” or alternative trait-based models (Jordano et al. 2006, Santamaría and Rodríguez-Gironés 2007), but should be evaluated against the contribution of information deficit due to limited sampling, representing a more simple and parsimonious explanation. The same effect applies to metrics such as generality, vulnerability, or linkage density of unweighted links, as they are directly related to C for a given network size (Table 1).

Nestedness

Nestedness (Table 1) in ecological interaction networks is commonly interpreted as specialization asymmetry: specialists (species with few links) interact with generalists (species with many links). Consequently, a nested pattern may suggest that reciprocal specialization, required for tight coevolution, is uncommon (Bascompte et al. 2003, Vázquez and Aizen 2004, Montoya et al. 2006). In most previous studies of natural networks, the associations were described as being highly nested, and significantly so in comparison with null models (Bascompte et al. 2003, Guimarães et

⁵ <http://www.nceas.ucsb.edu/interactionweb/>

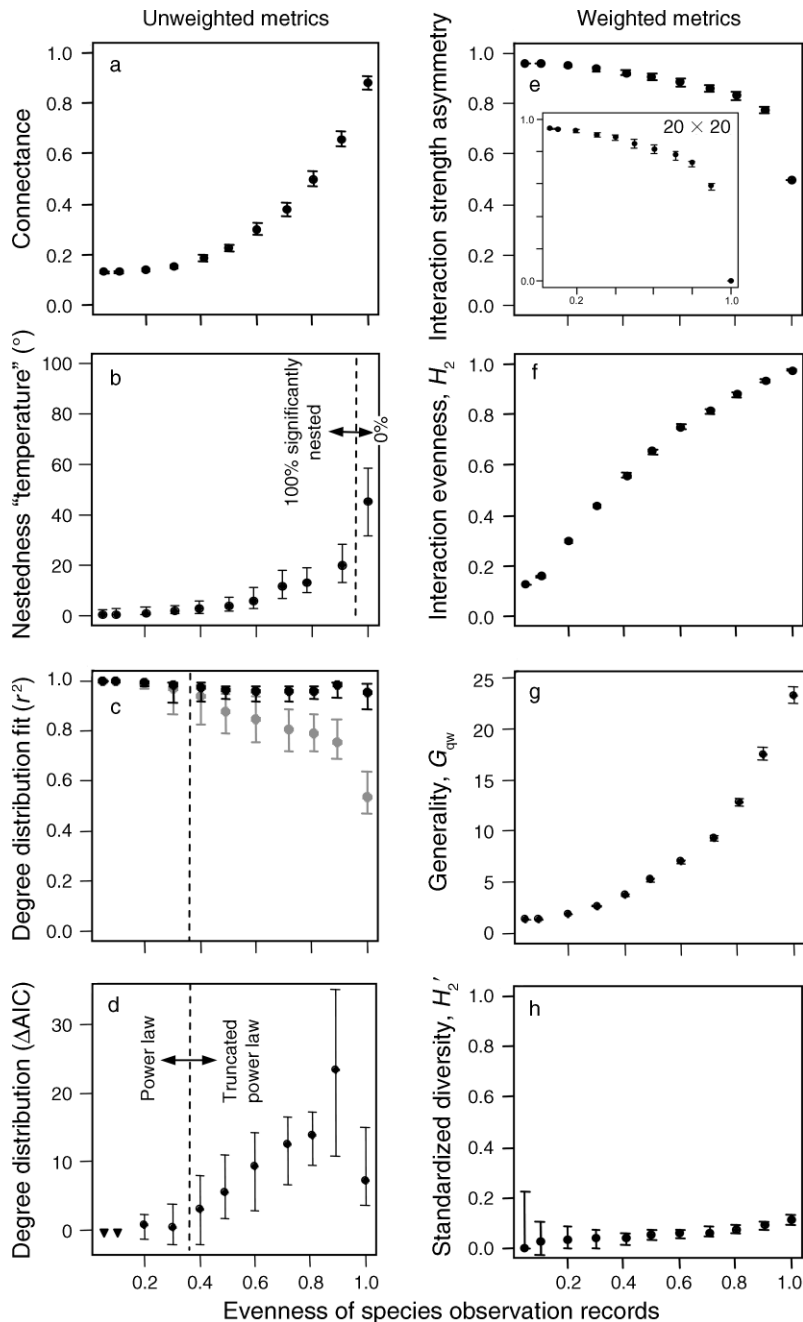


FIG. 4. Effect of the heterogeneity in species observation records ($\mathbf{R}_k$) on unweighted (a–d) and weighted (e–h) network metrics (see Table 1 for definitions). Simulated networks contain 30×10 species that interact randomly in the most generalized way, unconstrained by trait matching. Heterogeneity of $\mathbf{R}_k$ is displayed as Shannon evenness. For a given $\mathbf{R}_k$, mean with 95% CI for 100 networks is shown. The dashed line in (b) separates the range of $\mathbf{R}_k$ leading to significantly nested networks according to a null model used elsewhere (“CE” in Guimarães and Guimarães [2006]) in all cases (100%) vs. none (0%) ($n = 100$ networks for each $\mathbf{R}_k$). (c) The fit (r^2 value) of the power law (black circles) and truncated power law (gray circles) degree distribution, which correspond to (d) differences in Akaike Information Criterion (ΔAIC) between both distributions. For evenness values higher than the dashed line, the truncated power law provides a sufficiently better fit (mean $\Delta AIC > 2$). The inset in panel (e) shows the pattern for a squared network (20×20 species). In (g), generality is shown as a quantitative weighted (qw) mean for the party with 10 species.

al. 2006, 2007, Ollerton et al. 2007). However, a heterogeneous abundance distribution alone, which is typical for community samples, generates a nested pattern when partners are randomly associated (Fischer

and Lindenmayer 2002, Lewinsohn et al. 2006). Our analysis confirmed this effect for simulated neutral networks and additionally demonstrates that the evenness of observed species frequencies ($\mathbf{R}_k$) greatly

decreased nestedness: matrix temperatures increase with increasing evenness of $\mathbf{R}_k$ (Fig. 4b). Although a potential impact of the abundance distribution has been appreciated earlier (Jordano 1987), appropriate controls have been scarce, and nestedness has often been interpreted directly in terms of specialization asymmetry.

As opposed to the original null models used in community ecology and biogeography (Atmar and Patterson 1993), most studies of nestedness in ecological networks used null models that allowed a heterogeneous distribution of the number of links among species and were regarded as more conservative (Bascompte et al. 2003, Guimarães et al. 2006, 2007, Ollerton et al. 2007). Nestedness in many empirical networks was reported to be significantly greater than expected by those earlier null models. However, most of the networks generated by our null model also lead to “significant nestedness” in terms of previous null models, except randomized associations with a nearly perfectly even $\mathbf{R}_k$ (Fig. 4b). This may be largely due to conceptual differences in the null model algorithm (Fig. 3). Null models to evaluate nestedness in ecological networks have so far been based on the species degrees (unweighted links) rather than frequency totals (but see Vázquez and Aizen 2003, Stang et al. 2007); such marginal totals of species degrees are typically less uneven than $\mathbf{R}_k$.

Therefore, nestedness is a common pattern of empirical networks, but also of simulated networks based on completely generalized interactions, where $\mathbf{R}_k$ shapes the likelihood of interactions irrespective of trait matching (Fig. 3). If partners associate randomly, species are more likely to interact with common than with rare partners. Singletons or rarely observed species will be recorded interacting with common species, producing a nested pattern. This should be considered when comparing empirical data and interpreting nestedness as a biologically meaningful trait. For instance, Santamaría and Rodríguez-Gironés (2007) used nestedness in order to test various trait models in plant–pollinator interactions. These authors admit that null models with random interactions alone explain the variation in both nestedness and connectance over network size when the underlying frequency distribution was assumed to be log-normal, but downplay the value of null models in general to describe plant–pollinator interactions realistically. However, we suggest that if sampling limitation under neutral conditions alone explains a pattern, a more complex trait model is not required as alternative explanation (parsimony principle). An insightful comparison between the explanatory powers of morphological trait models vs. neutral models in plant–pollinator networks has recently been performed by Stang et al. (2007).

Degree distribution

In many ecological networks sampled so far, few species have many links, while many species have only

few links, although this pattern may vary across different networks and food webs. The resulting degree distribution often decays more strongly than in “scale-free” networks and can be better described by a truncated power law function (Table 1). This particular topology characterizes ecological networks in relation to some other types of networks (Dunne et al. 2002a, Jordano et al. 2003, Montoya et al. 2006). Ecological traits of interacting partners have been suggested to shape this particular degree distribution, such as the non-matching of species attributes that may prevent interactions between species (“forbidden links”; Jordano et al. 2003). However, networks with completely generalized interactions can also be very well characterized by truncated power law functions when observed total frequencies of different species are sufficiently uneven (often with $r^2 \approx 1.0$) (Fig. 4c). This corresponds to an earlier notion that the degree distribution can be generated by a random interaction model (Vázquez 2005). The heterogeneity of $\mathbf{R}_k$ has a strong influence on the fit of the power law distribution, and the truncated power law model provides a sufficiently better fit (considering AIC) over a large range of $\mathbf{R}_k$ except for networks with a very low evenness (Fig. 4d).

Interaction strength

Quantitative interaction strength in interaction networks (b_{ij}) is often defined as the proportion of the interactions i with a specific partner j of the total interactions of this species i (Table 1) (“dependence” sensu Jordano 1987, Bascompte et al. 2006). Thus, when a plant is visited solely by a single pollinator species, its interaction strength with that pollinator is assumed to be 1.0, while lower values indicate that this pollinator contributes only a smaller fraction of the visits. Asymmetric interaction strengths have been interpreted as a quantitative cousin of qualitative specialization asymmetries and nestedness: when a pollinator is highly dependent on a plant, this plant often exhibits low dependence on the pollinator in turn, and this asymmetric dependence may enhance community stability (Bascompte et al. 2006; but see comment by Holland et al. [2006]). However, the same argument applies as for nestedness above: when assuming neutral encounters among interacting species suffices to reproduce observed asymmetric interaction strengths (based on $\mathbf{R}_k$), it is not imperative that trait matching is generating such asymmetry.

Indeed, simulated random networks based on a heterogeneous distribution of interaction totals (low evenness of $\mathbf{R}_k$) usually produce high asymmetries in interaction strength between species. With an increasingly homogenous abundance distribution, average asymmetries decline (Fig. 4e). For squared randomized networks ($I = J$), a uniform interaction frequency (perfectly even $\mathbf{R}_k$) leads to completely symmetric interaction strengths (see inset in Fig. 4e). An increased imbalance in the number of associated partner species (I

$\neq J$, network asymmetry) generates higher interaction strength asymmetries even for uniform abundances (Fig. 4e). This confirms that both sampling intensity and network asymmetry (Blüthgen et al. 2007) shape the average interaction strength asymmetry for neutral interactions without any effect of trait matching. Correspondingly, Vázquez et al. (2007) showed that the predominance of interaction strength asymmetries in many mutualistic and antagonistic networks may reflect their underlying abundance distribution.

Interaction strengths are highly constrained for infrequently recorded species. Most notably, an interaction by a species with a single observation will automatically be assigned an interaction strength of 1.0, two observation records will either lead to a value of 1.0 if they occur on the same link or 0.5 for each of two links, and so forth. Thus, rarely observed species will have, on average, higher interaction strengths than frequently sampled species. In a given network, even the majority of high interaction strengths assigned (e.g., values ≥ 0.5) may involve species with only one or two total observation records (including those networks analyzed by Bascompte et al. 2006). However, because high values of interaction strength b_{ij} often correspond to poorly sampled and infrequent species, these may not reflect particularly “strong” interactions as implied by Bascompte et al. (2006). If there is an inverse relationship between sampling intensity and interaction strength, and if rarely observed species typically associate with commonly observed partners, this combination of features may produce interaction strength asymmetry in networks. In order to account for this bias in interaction strength for low number of observations in species i (n_i), interaction strength could be standardized for $n_i > 1$ as

$$b'_{ij} = (b_{ij} - b_{\min}) / (1 - b_{\min})$$

where $b_{\min} = 1/n_i$ excluding all cases where $n_i = 1$. Such standardized b'_{ij} vary between 0 and 1 irrespective of n_i and may arguably underestimate the real interaction strength, but may provide a useful way to test for effects of potential bias due to the information deficit under neutral conditions. Incorporating the modified equation for b'_{ij} and the condition $n_i > 1$ in our simulation of randomized networks often leads to more moderate values of interaction strength asymmetries, but does not fully compensate for their dependence on $\mathbf{R}_k$ evenness (see Appendix). Alternatively, interaction strengths may be weighted according to the number of observations, as the credibility given to interaction strengths calculated for each species increases with n_i .

Interaction diversity, evenness, and generality

Bersier et al. (2002) suggested several metrics based on Shannon diversity (Table 1) to describe the heterogeneity in mass flows in food webs, which can be transferred to characterize the diversity of associations based on interaction frequencies. Correspondingly, other authors

suggested Simpson's diversity metric to characterize generalization in plant–pollinator networks (Sahli and Conner 2006). In an extensive study of bee–parasite associations, Tylianakis et al. (2007) showed that Shannon diversity and evenness (i.e., Shannon diversity/log species richness) of interactions decreased as a function of habitat disturbance. This pattern depicts the heterogeneity of the interactions between trophic levels, but not necessarily changes in specialization. Not surprisingly, in our simulation interaction evenness (and consequently interaction diversity) is almost perfectly correlated with the evenness of the total observation records ($\mathbf{R}_k$) (Fig. 4f). Consequently, a pattern reported for empirical networks (e.g., Tylianakis et al. 2007) may be driven by a decrease in evenness of the host or resource abundances alone, even when preferences of consumers or parasites are unchanged. The impact of $\mathbf{R}_k$ is equally important for other metrics derived from diversity measures such as generality or vulnerability (Fig. 4g).

In order to characterize specialization, Hurlbert (1978) emphasized in the analogous context of niche breadth that appropriate measures should not only take the distribution of consumers into account, but also control for variation in resource availability. This concept is implemented in the following metrics (H'_2 and d'_i) that are based on the Shannon diversity of interactions as well, but are standardized against a null expectation.

Quantitative metrics that control for $\mathbf{R}_k$

The standardized two-dimensional Shannon diversity H'_2 and the related standardized Kullback–Leibler distance d'_i describe the degree of mutual specialization among two parties in the entire network and for each species, respectively (Table 1; Blüthgen et al. 2006). These metrics calculate the deviation of realized associations from the null expectation based on the frequency distribution of interaction totals $\mathbf{R}_k$. Consequently, in randomized associations H'_2 and d'_i remain near zero and are largely unaffected by the evenness of the observation records (Fig. 4f; Blüthgen et al. 2006), and these metrics are robust against variable species frequencies or biased sampling intensity. Therefore, any deviation from this null expectation may be driven by processes at the interspecific level such as trait matching. For instance, a higher number of interactions recorded for a certain link than predicted by the null model may suggest a preference of the animal species for this resource, whereas a lower frequency may be indicative of a morphological barrier, defenses, or other inhibitory processes (Fig. 1). Without reference to the null model, however, it may be difficult to distinguish whether unobserved or underrepresented links are due to limited sampling or due to biologically meaningful restrictions (compare scenarios b and c in Fig. 1). In other words, failure to record a link between two rarely observed partners may not be surprising, even when an obvious

morphological barrier is lacking, whereas an unobserved link between a common consumer and a common resource more vigorously calls for a biological interpretation.

The null model relates to the concept of nestedness in a quantitative way, implying that each species opportunistically visits its partners in the same proportions as the other species do, so that all species conform to the same overall $\mathbf{R}_k$. Consequently, interactions of each species are a subset of the interactions of more frequently observed species. A quantitatively “nested” network based on weighted links looks like the one displayed in Fig. 2a. For a perfectly nested matrix, $H'_2 = 0$. Therefore, generalized interactions generate a pattern of quantitative nestedness. The pattern opposite to nestedness would be a scenario with strongly symmetric (reciprocal) specialization between partners, i.e., highly mutually exclusive interactions. This corresponds to an interaction matrix in which mainly the diagonal elements are realized when species are sorted by their frequency (Fig. 2d), a scenario that yields $H'_2 = 1$. Consequently, H'_2 is an index that describes specialization in terms of exclusiveness (or reciprocal specialization), where $H'_2 = 0$ represents the most nested scenario possible given the overall frequency distribution, and $H'_2 = 1$ the highest possible deviation from this nested assemblage. In contrast, note that the concept of “anti-nestedness” in unweighted networks is controversial and has been used for several distinct network characteristics (Almeida-Neto et al. 2007). Conventional nestedness analyses based on unweighted links identify species that particularly deviate from the expectations based on the most nested pattern, shown by high “idiosyncratic temperatures” (Atmar and Patterson 1993). Such idiosyncratic temperatures in unweighted networks have their counterpart in high d'_i values in networks with weighted links. For species that exclusively interact with their partners (i.e., reciprocal specialization) and have thus nothing in common with their putative competitors, d'_i approaches 1. Species with high d'_i are, e.g., bellflowers (*Campanula* spp.) visited by a particularly unconventional spectrum of bees (J. Fründ, unpublished data), suggesting a symmetric mutual dependence of the partners. These cases of reciprocal specialization may represent targets that deserve particular attention for conservation. In contrast, low d'_i may suggest a higher ecological redundancy with members of the same guild.

Most mutualistic interaction networks show a higher exclusiveness (H'_2) than expected by random interactions based on the marginal totals ($\mathbf{R}_k$), thus they are significantly less nested quantitatively than expected by the null model (Blüthgen et al. 2007). This significant exclusiveness is also apparent for antagonistic networks. For instance, in fish–parasite associations (seven networks from Vázquez et al. [2005]), H'_2 ranged between 0.34 and 0.74, and all networks were significantly more exclusive than expected by the null model (all $P < 0.001$). Therefore, empirical networks as well as ran-

domized associations with a skewed $\mathbf{R}_k$ may appear nested when analyzed as presence–absence matrices (Fig. 3), whereas our analysis based on weighted links reveals that empirical networks are usually not nested and are even more exclusive than expected by chance (Fig. 2).

Differences in the degree of specialization (H'_2) among different network types may suggest structural differences in phenotypic matching of partners, morphological or spatiotemporal constraints, and active behavioral choices triggered by species preferences and avoidance. High specialization also suggests a higher potential of coevolutionary processes to shape interaction networks than implied by qualitative nestedness (Thompson 2005). Antagonistic associations, e.g., host–parasite or plant–herbivore systems, are strongly driven by host defenses and subsequent consumer offenses, and this evolutionary arms race may promote a high exclusiveness of interactions due to strong limitations and barriers at the level of trait matching. Among mutualistic systems, benefits from specialization may differ. For instance, the relatively high exclusiveness in pollinator networks may reflect the plants’ interest in reliable pollen vectors that transfer pollen to conspecific flowers, compared to low H'_2 in seed dispersal networks (Blüthgen et al. 2007).

In terms of community stability, a smaller overlap between species of the same guild in terms of interaction frequencies suggests a lower redundancy of species, and thus a higher potential of negative effects of a population decline of a host or resource species on its associated consumer species. In addition, high levels of exclusiveness in mutualistic networks may indicate a pronounced reciprocal dependence between interacting partners.

CONCLUSIONS

In this study we simulated associations at the highest possible level of generalization, i.e., no limitation by trait matching. In this null model scenario, every species i can associate with every potential partner j , only constrained by the total observations per species ($\mathbf{R}_k$ vectors). Failure to observe a link between i and j , or a relatively low interaction strength between i and j , is therefore only an effect of insufficient sampling, not of trait matching. In contrast to the expected maximum degree of generalization in all simulations, most metrics are fundamentally affected by the evenness of $\mathbf{R}_k$ alone, without intentionally changing the generalized character of associations between partners. The impact of $\mathbf{R}_k$ is severe for both unweighted and weighted interaction network metrics, and represents a realistic constraint in empirical network analysis, considering the typical sampling limitation in ecological data, and given that many species in ecological samples are rare or rarely observed (see Plate 1).

Conventional network metrics render all links equally likely, irrespective of whether the absence or weakness of

a link results from mismatch of traits or is solely due to limited information. Therefore, any assessment of the determinants structuring networks must consider to what extent variation in number and distribution of observed interactions may be solely enforced by limited sampling intensity. The standardized indices H'_2 and d'_i (Blüthgen et al. 2006) represent a solution to compare networks directly by controlling for variable observation records and abundances under neutral conditions, and are thus more conclusive about different structural forces at the level of trait matching. The comparison with the expectation by the neutral model facilitates an interpretation of the presence, absence, or strength of a link, particularly when external data about ecological barriers (e.g., as in Stang et al. 2007) are lacking. An absence of a link in a real network that was also unexpected by the respective null model may represent an information deficit (Fig. 1c). In contrast, other gaps in the realized associations that are expected to occur frequently under neutral conditions may suggest an important underlying mechanism such as a morphological or physiological barrier or phenological mismatch (Fig. 1b). Without reference to the null model, all links or interaction strengths are weighted equally.

The need to control for $\mathbf{R}_k$ to evaluate trait matching does not imply that uncontrolled metrics are meaningless. For instance, in order to describe energy or mass flows between trophic levels in food webs (Bersier et al. 2002) or heterogeneity of interactions (Tylianakis et al. 2007), a measurement of the diversity of links (e.g., unstandardized Shannon diversity) may be useful. This is also true if different impacts of consumers on their host are to be evaluated and increase with their overall frequency (e.g., pollination, predation, or herbivory) (e.g., Sahli and Conner 2006). Asymmetric interaction strengths may describe an important pattern in realized associations if the recorded frequencies are representative for the species' overall activity. Depending on the scope of the analysis, variation in $\mathbf{R}_k$ may represent a structural property or emergent feature not to be deducted from the overall pattern. However, the uncontrolled metrics are problematic in their interpretation of fundamental specialization among associated species, i.e., a direct conclusion about processes that act at the interguild level of trait matching. For each area of ecological analysis it is therefore crucial to choose appropriate metrics in order to avoid flawed or misleading interpretation of network data. Use of appropriate standardizations and/or comparisons with null models is vital to understand and correctly interpret the structuring elements of a pattern while removing effects of limited or biased sampling. Rarefaction techniques additionally help to unravel the contribution of sampling intensity for quantifying specialization and network patterns (Gotelli and Colwell 2001, Herrera 2005). Most importantly, the frequency distribution of the species observations ($\mathbf{R}_k$) itself and its underlying determinants such as the abundance distribution

(McGill et al. 2007), merits a more intensive focus in future investigations.

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APPENDIX

Effect of the heterogeneity in species observation records on network metrics for an alternative parameter setting (*Ecological Archives* E089-194-A1).

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Appendix A. Effect of the heterogeneity in species observation records on network metrics alternative parameter setting.

Here we provide the results of the simulation for a different parameter setting (Fig. A1). Generated networks contained $I \times J = 20 \times 20$ species and a total of $m = 400$ interactions, equivalent to an average of one interaction per cell ($m / [IJ] = 1$). The most liberal alternative null model was used, where the total observation frequencies $\mathbf{R}_k$ defined the probability of an interaction but was not fixed unlike in the Patefield algorithm, and the condition to maintain one link per species was not implemented, partly resulting in a reduced number of species.

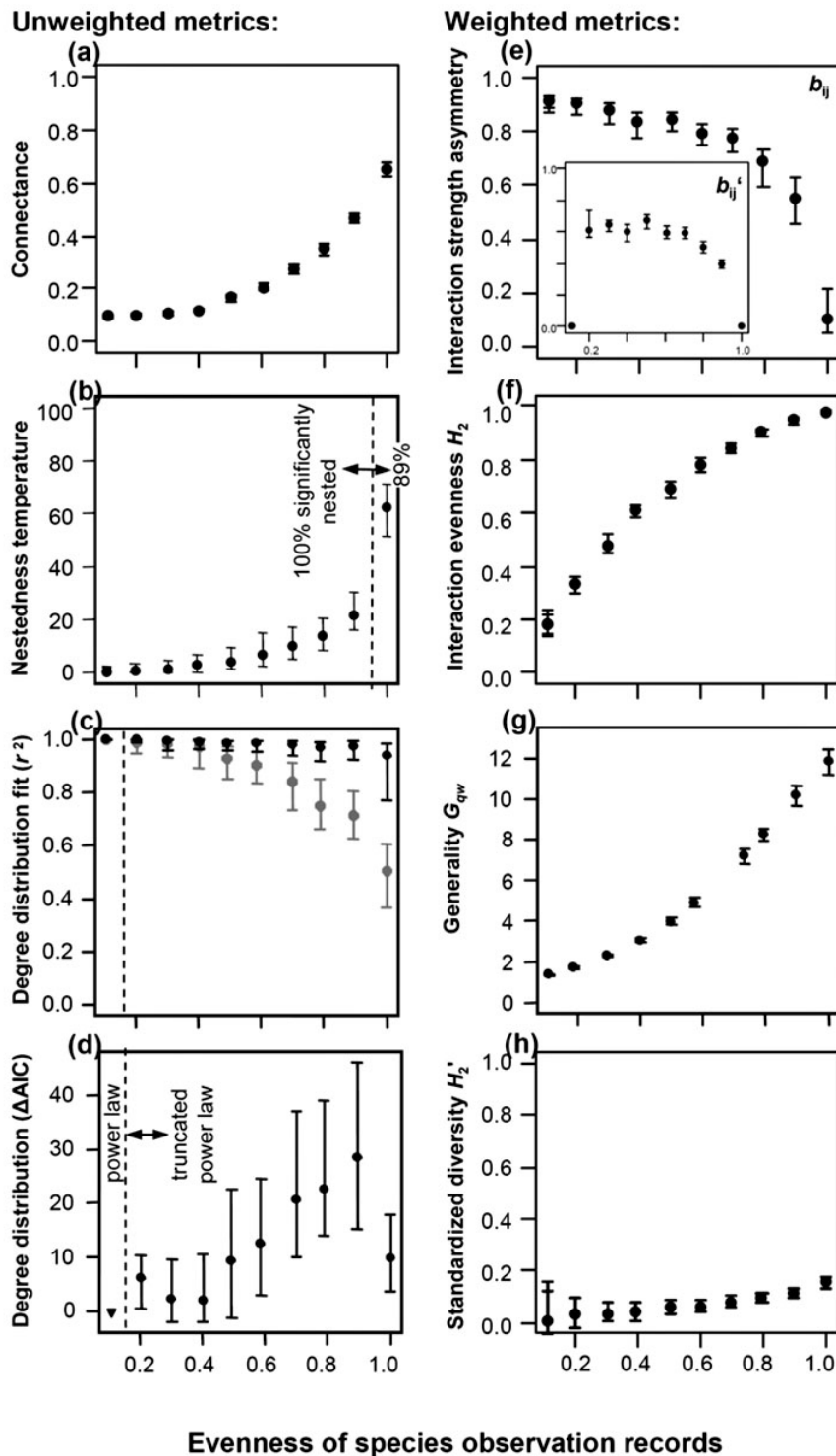


FIG. A1. Effect of the heterogeneity in species observation records ($\mathbf{R}_k$) on unweighted (a–d) and weighted (e–h) network metrics (mean $\pm$ 95% C.I. for 10 networks). Heterogeneity of $\mathbf{R}_k$ displayed as Shannon evenness. Dashed line in (b) separates the range of $\mathbf{R}_k$ leading to significantly nested networks according to null model ‘CE’ in all cases (100%) versus a lower proportion (89%) ($n = 100$ networks for each $\mathbf{R}_k$). In (c), the fit (r^2 value) of the power law (black circles) and truncated power law (gray circles) degree distribution shown, which correspond to differences in ΔAIC between both distributions (d). For evenness values higher than the dashed line, the truncated power law provides a sufficiently better fit (mean $\Delta AIC > 2$). Inlet in (e) shows the pattern for networks based on the Patefield algorithm and a modified equation for interaction strength, $b_{ij}^{\epsilon} = (b_{ij} - b_{\min}) / (1 - b_{\min})$, where $b_{\min} = 1/n_i$; all cases where $n_i = 1$ have been excluded. For the other null models, b_{ij}^{ϵ} showed a similar decrease over $\mathbf{R}_k$ evenness as the unstandardized b_{ij} .